

Dimensionality Reduction MCQ (Version : 2)

TEST

● **Correct Answer**

🕒 Answered in 3.73 Minutes

Question 1/15

Principal Component Analysis is a linear dimensionality reduction technique.

☒ True

☐ False

Explanation:

PCA (Principal Component Analysis) is a linear dimensionality reduction technique. It transforms data into a new coordinate system where axes represent principal components, linear combinations of original features.

Question 2/15

Before performing PCA, what should ideally be done to the data set?

☐ normalise variables

☒ standardise variables

☐ encode variables

☐ categorise variables

Explanation:

The correct answer, standardising variables

(subtracting mean, dividing by standard deviation), is crucial for PCA to ensure equal contribution of variables.

Normalising variables is incorrect because normalisation may not preserve variance.

Categorising variables is incorrect because categorisation isn't a preprocessing step for PCA.

Encoding variables is incorrect because encoding is used for categorical variables, not specifically for PCA.

Question 3/15

What can be said of the eigenvector with the largest eigenvalue, in terms of its relation to the principal components of a dataset?



It corresponds to the direction with the largest variance in the data



It equals the variance in the first principal component



It equals the variance in the last principal component



It corresponds to the direction with the least variance in the data

Explanation:

- It corresponds to the direction with the largest variance in the data: The eigenvector with the largest eigenvalue represents the direction of maximum variance in the data, corresponding to the first principal component.
- It corresponds to the direction with the least variance in the data: is incorrect because the largest eigenvalue corresponds to maximum variance.
- It equals the variance in the first principal component: is incorrect because it represents variance, not specifically in the first principal component.

- It equals the variance in the last principal component: is incorrect because it doesn't correspond to the largest eigenvalue.

Question 4/15

Which of the following is NOT a method for dimensionality reduction?

☐ Principal Component Analysis

☒ k-means clustering

☐ Multidimensional Scaling

☐ t-SNE

Explanation:

- k-means clustering is a clustering algorithm, not a dimensionality reduction technique.

- The other options are dimensionality reduction techniques.

Question 5/15

True or false: t-SNE is a linear dimensionality reduction technique.

☒ False

☐ True

Explanation:

t-SNE (t-Distributed Stochastic Neighbour Embedding) is a non-linear dimensionality reduction technique

Question 6/15

Which of the following is not true regarding dimensionality reduction:

☐ it decreases the interpretability of models

☒ it always improves performance of clustering algorithms

☐ it decreases the computational time for training models

☐ it can be used to aid data visualization

Explanation:

- It always improves performance of clustering algorithms: Correct because dimensionality reduction doesn't always guarantee improved clustering performance; it depends on the data and method used.
- It decreases the computational time for training models: Incorrect because dimensionality reduction can reduce computational time.
- It decreases the interpretability of models: Incorrect because it may increase interpretability by focusing on most important features.
- It can be used to aid data visualisation: Incorrect because dimensionality reduction techniques like PCA and t-SNE are used for data visualisation.

Question 7/15

One of the hyperparameters available for tuning in t-SNE is:

☐ complexity

☒ perplexity

☐ irregularity

☐ duplicity

Explanation:

- Perplexity is a hyperparameter in t-SNE controlling the number of close neighbours each point considers.
- Complexity, irregularity, and duplicity are incorrect because they are not hyperparameters of t-SNE.

Question 8/15

Which of the following two dimensionality reduction techniques preserves distances in lower dimensions?

☒ Metric Multidimensional scaling

☐ t-SNE

Explanation:

- Multidimensional Scaling (MDS) preserves distances between points in lower-dimensional space.
- t-SNE is incorrect because it does not preserve distances.

Question 9/15

It is not necessary to have a target variable for applying dimensionality reduction algorithms.

☐ False

☒ True

Explanation:

Dimensionality reduction algorithms like PCA and t-SNE do not require target variables.

Question 10/15

One of the most popularly used dimensionality reduction algorithm is Principal Component Analysis (PCA). Which of the following is/are true about PCA?

- i. PCA is an unsupervised method
- ii. PCA seeks to identify vectors in the original feature space that correspond to directions where the data exhibit the highest variance.
- iii. The maximum number of principal components $\leq$ number of features
- iv. All principal components are orthogonal to each other



i, ii, iii and iv



i, ii, and iv



i, iii and iv



i, ii and iii

Explanation:

Given that all four statements are true, the correct choice is "i, ii, iii and iv". This means that PCA is indeed an unsupervised method, it searches for directions with the largest variance, the maximum number of principal components is limited by the number of features, and all principal components are orthogonal to each other.

Question 11/15

In which of the following scenarios is t-SNE better to use than PCA for dimensionality reduction while working on a local machine with minimal computational power?



Data set with 10,000 entries and 8 features



Data set with 1 Million entries and 300 features



Data set with 100000 entries and 310 features



Data set with 10,000 entries and 200 features

Explanation:

- Data set with 10,000 entries and 8 features: correct because t-SNE is better suited for smaller datasets with fewer features like the one described.
- The other options are incorrect because they describe larger datasets where PCA might be more suitable due to its lower computational complexity.

Question 12/15

In t-SNE algorithm, which of the following hyper parameters can be tuned?



Number of dimensions, smooth measure and max iterations



Maximum number of iterations



Number of dimensions



Smooth measure of effective number of neighbours

Explanation:

- Number of dimensions, perplexity, and max iterations: Correct. The number of dimensions, smooth measure (perplexity), and maximum iterations are tunable hyperparameters in t-SNE.
- The other options are incorrect because they do not accurately list all tunable hyperparameters of t-SNE.

Question 13/15

Which of the following statement is correct for t-SNE and PCA?

☐ t-SNE and PCA both are linear

☐ t-SNE is linear whereas PCA is non-linear

☐ t-SNE and PCA both are nonlinear

☒ t-SNE is nonlinear whereas PCA is linear

Explanation:

t-SNE is nonlinear, while PCA is linear. Because t-SNE preserves local structures in the data, while PCA finds orthogonal linear combinations of features.

Question 14/15

What might happen when eigenvalues are roughly equal?

☐ All eigenvectors will be the same

☒ PCA will perform poorly

☐ PCA will perform optimally

☐ The number of dimensions in the data will increase

Explanation:

When eigenvalues are roughly equal, the dimensionality reduction might result in a larger loss of information than if there had been components with larger differences in eigenvalues.

Question 15/15

An embedding is a representation of a vector in a different feature space.



True



False

Explanation:

True. An embedding is indeed a representation of a vector in a different feature space, often achieved through dimensionality reduction techniques.